Taking Aztec Global Freight From Reactive Response to Disruption Intelligence

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Aztec Global Freight records what happens to every shipment, but it does not yet use those records to decide what to do before a shipment fails. This paper proposes a disruption intelligence framework that follows the specify, assess, and mitigate logic of disruption risk management (Kleindorfer & Saad, 2005). The provided datasets (orders, international legs, domestic legs, events, and nodes) were analyzed in Excel and in Python with the pandas and scikit-learn libraries. Results of that analysis are reported in the past tense. The database, application programming interface (API), dashboard, and operating controls are proposed and have not been built. These are described in the future tense.

The following results shape recommendations… First, 1,265 orders (42.2%) missed their promised delivery time, and half of those misses (632 orders, all imports) were set at booking because the wait before departure alone exceeded the promise buffer, so they would have been late even with no transit delay. Second, after the tariff increase took effect, 470 shipments from China paid $14.01 million in tariffs, $6.19 million more than the old rate would have charged. Of those shipments, 18 could have avoided the increase by switching to air when it was announced. Third, fields that a predictive system depends on, such as event links and tariff dates, disagree with Event_Calendar and must be reconciled before they are used as model inputs (Hazen et al., 2014).

The analysis rests on the following assumptions: (a) the provided data are accurate; (b) planned transit times and the booked departure date are known when an order is placed; (c) freight cost = intl_total_cost_usd − tariff_cost_usd; (d) a service level agreement (SLA) violation is any delivery after promised_delivery_time; and (e) any value the data do not provide, such as the cost of a missed delivery, is stated as an assumption where it is used.

Question 1: Information Requirements

Multistream Event Fusion (1.1)

In the proposed system, each event from the weather, seismic, conflict, geopolitics, and operations feeds will be normalized into a common Event_Calendar record and fused at three levels, corresponding to Levels 1 to 3 of the Joint Directors of Laboratories data fusion model (Hall & Llinas, 1997). At the object level, duplicate reports of the same event will be merged when they match on event_type, geo_tag, and window. At the situation level, each geo_tag will be resolved to Node_Lookup entries (e.g., LA-LB to two ports of entry; Red Sea to Asia–East Coast lanes), and a leg will be marked exposed when the affected area contains one of its nodes or lanes and the event window overlaps the leg's planned window. At the impact level, exposures will become order features: active event count, maximum severity, and a compound flag set when both legs are exposed.

This layer is needed because the existing event tags cannot be reused. Of the 516 international legs carrying a trigger_event_id, only 11% to 34% overlapped their event's window compared with 2% to 6% of the 568 tagged domestic legs. EVT001, a Shanghai typhoon, was attached to all seven origin ports, and only 12 of its 95 legs sailed from Shanghai. Recomputed exposure did not raise the disruption rate (international: 28.3% vs. 30.9%, $p = .37$; domestic: 20.6% vs. 17.8%, $p = .47$). The fusion layer will still be built because the alerts and network map depend on it. Its features will enter the model only after they prove predictive on later data.

Modeling Tariff Announcement and Effective Dates (1.2)

A tariff needs two linked Event_Calendar records. The announcement (EVT005) which opens a decision window in which Aztec can act early, and the effective date (EVT006) which fixes liability for every shipment that arrives after it. Because duty is assessed at entry, the

announcement flag on intl_departure_time and the effective flag should key on intl_arrival_port_time . The announcement flag marks departures from China between April 20 and May 14, 2025, and the effective flag marks arrivals from China from May 17, when the tariff rate, weighted by order value, rose from 14.4% to 25.9%. Because the 26 days of notice equal the median planned ocean transit from China, 33 of the 37 orders that sailed in the window arrived after the effective date. Tariffs pass almost fully into importer prices (Amiti et al., 2019), and uncertainty shapes sourcing (Handley & Limão, 2017).

The sources disagree: Event_Calendar dates EVT005 to June 16 and EVT006 to July 17, two months after the rate change, and 189 flagged arrivals follow EVT006's October 15 end. In the proposed system, flags will therefore be derived from Event_Calendar by rule and reconciled against the tariff charged on every load. This analysis used the leg flags, which match the tariff rates charged.

Operational Definition of Disruption Ground Truth (1.3)

An order is labeled disrupted when intl_delay_days is ≥ 2 days or domestic_delay_hours is ≥ 12 hours. The thresholds follow from the promise buffer: promised_delivery_time minus order_time minus planned transit time. For imports, the buffer remaining after the wait before departure had a median of 18 hours and was under 48 hours for 86.1% of orders. The miss rate was 30.3% with no international delay, 64.9% at one day, 91.1% at two, and 100% beyond. Domestic delays were nearly all or nothing (2,432 orders with none, 31 under 12 hours, 537 at 12 or more). Twelve hours was the smallest buffer in the extract, and a delay of 12 hours or more raised the miss rate from 34.5% to 75.8%.

The selected rule labeled 980 orders (32.7%), 85.7% of which missed the SLA. It captured 94.7% of the 887 misses that had any transit delay and agreed with any_disruption_flag

on 93.4% of orders. As Table 1 shows, an international trigger at 1 day caught 45 more misses but added 25 false positives, a trigger at 5 days missed 611 failures, and a domestic trigger at 24 hours cut false positives from 140 to 80 but missed 36 more. The remaining 378 late orders had no transit delay and were all among the 632 with infeasible promises, so SLA failure was modeled as a separate outcome.

Table 1

Ground Truth Threshold Sensitivity Across Orders

Rule (international / domestic)	Labeled	Precision	Recall	False positives	Missed failures
≥ 1 day / ≥ 12 h	1,050	84.3%	70.0%	165	380
≥ 2 days / ≥ 12 h (selected)	980	85.7%	66.4%	140	425
≥ 2 days / ≥ 24 h	884	91.0%	63.6%	80	461
≥ 5 days / ≥ 12 h	784	83.4%	51.7%	130	611

Note. Precision is the share of labeled orders that missed the SLA, and recall is the share of the 1,265 misses labeled.

Integrating International and Domestic Layers (1.4)

Each order has at most one International_Leg row (imports only) and exactly one Domestic_Leg row, linked through order_id. The four node columns reference Node_Lookup.node_name, and trigger_event_id is a nullable foreign key to Event_Calendar. The extract passed every key and orphan check, and the handoff is explicit: on every import, (us_port_of_entry = domestic_origin_node) and (domestic_ship_time = intl_arrival_port_time). Across all 3,000 joined orders, an order was late exactly when the wait before departure plus international and domestic delay exceeded its buffer, so domestic performance cannot be judged apart from international delays. One gap remains, the 412 customs holds have no timestamp of their own. Every held shipment also had an international delay long enough to contain its hold,

so hold time appears to be folded into intl_delay_days, and the data cannot separate customs time from transit time.

Data Governance Requirements (1.5)

Timestamps will be stored in Coordinated Universal Time (UTC) under ISO 8601 (International Organization for Standardization [ISO], 2019). Country fields already use ISO 3166-1 alpha-2 codes (ISO, 2020), and ports will carry UN/LOCODE codes such as CNSHA alongside node_name (United Nations Economic Commission for Europe, n.d.). Carriers need a master list of permitted modes because every carrier appeared on every mode, including DHL Air on 380 ocean legs and RailLink on 461 parcel and truck legs. Missingness needs one policy because disruption fields held the text “None” while trigger_event_id was left blank. Under the missingness policy, an absent value will always be stored as NULL, and each NULL will be either expected by design, such as no event on an undisrupted leg, or logged as an error, such as the 370 tariff disruptions with no event link. Deduplication will keep the latest record per order_id and leg and merge event reports with matching event_type, geo_tag, and window. The data contained no duplicate keys. Each rule will have a data owner (Khatri & Brown, 2010) and will be measured on quality dimensions defined by its users (Wang & Strong, 1996).

Validating the Requirements

Stakeholders will confirm each requirement in a traceability matrix: operations the disruption thresholds, trade compliance the tariff dates, finance the miss cost, and customer service the SLA definition. Empirically, every rule will run as a test on each load. Nine structural checks already passed at 100% on the extract, while the checks on event links and tariff dates failed and will be reconciled before use. The ground truth rule will be rechecked against SLA outcomes, and models will be judged on later periods, as in this analysis (Hevner et al., 2004).

Question 2: Technology-Supported Architecture

Integrating the Schema's Five Tables (2.1)

The system described below has not been built. Only the models in Section 2.2 were trained and tested. The five tables will remain the system of record and will never be altered, and everything else will be a view or log keyed to them. A local PostgreSQL database will hold the tables, following the relational model (Codd, 1970). Primary keys will be Orders.order_id, International_Leg.order_id, Domestic_Leg.order_id, Event_Calendar.event_id, and Node_Lookup.node_name. Foreign keys will link both legs to Orders, the four node columns (origin_port, us_port_of_entry, domestic_origin_node, and customer_location) to Node_Lookup, and trigger_event_id to Event_Calendar. Data will move through three layers: stage, validate, and join (through a v_order_journey view). The view will match the join used in this analysis, with a left join from Orders to International_Leg and an inner join to Domestic_Leg, and will add computed fields such as promise buffer, wait before departure, and event exposure. Scores and alert decisions will be written to logs that can be added to but never edited.

Operationalizing the Predictive Model (2.2)

Two models match the two moments when Aztec can still act. The booking model (T0) used fields known at order entry: lanes, modes, carriers, planned transit, promise buffer, planned wait before departure, tariff flags, and event exposure. The model at port arrival (T1) added intl_delay_days and customs holds. Both predicted SLA failure with gradient boosting (Chen & Guestrin, 2016; Friedman, 2001) and were compared against a logistic regression benchmark. They were trained on January to August 2025 orders (2,007) and tested on September to December (993). On the test months, T0 reached an area under the receiver operating characteristic curve (ROC-AUC) of .849 and T1 .983, where .5 is chance and 1.0 is perfect

(logistic regression: .835 and .980). Without the buffer and wait fields, T0 fell to .687, so promise feasibility was the strongest early signal. Because planners judge alerts by how many flagged orders truly fail, precision and recall are reported alongside ROC-AUC (Fawcett, 2006; Saito & Rehmsmeier, 2015). At a flagging threshold of 50% predicted miss probability, T1 flagged 386 orders with 99.7% precision and 92.5% recall, and T0 flagged 316 with 78.8% precision and 59.9% recall. The result is consistent with Brintrup et al. (2020), who predicted supplier delivery disruptions from a manufacturer's own order records.

In operation, scores will become actions through a prescriptive cost rule (Lepenioti et al., 2020). An action is taken when

$$\text{score} \times \text{recovery probability} \times \text{cost of a miss} > \text{cost of the action}$$

The schema holds no penalty field, so this analysis assumes a miss costs $1,000 in service credits, expediting, and customer service time. For a recoverable shortfall (recovery probability of 1), the score at which acting breaks even is the action's cost divided by $1,000: .06 for Parcel_Standard to Parcel_Expedited ($63), .22 for Intermodal to full truckload (FTL; $219), and .52 for less than truckload (LTL) to FTL ($517). Each action therefore has its own threshold. Models will be retrained monthly and monitored for drift.

API Endpoints (2.3)

A local FastAPI service will expose the models to other systems (Table 2).

Table 2

Proposed API Endpoints

Endpoint	Input	Output
POST /predict	order_id, decision point (T0 or T1)	Miss probability, risk band, top three drivers, model and data versions, recommended action, its cost, and the threshold applied

GET /alerts	Optional filters: status, region, action, min_saving	Open alerts ranked by expected avoidable cost
POST /alerts/{id}/decision	Decision (approve or override), reason_code	An entry in the decision log
GET /scenario/tariff	event_id and action_date	Locked, addressable, and avoided exposure (the Table 3 calculation)

Streamlit Decision Support (2.4)

The proposed Streamlit dashboard will run in a web browser, draw on the API in Section 2.3, and open on a single overview screen that puts exceptions first (Few, 2006). Each further view is matched to the role that uses it:

- Overview: share of orders on time, open alerts, tariff exposure, and disruption rate against the pilot targets in Section 3.4.
- Alert queue (planners): orders ranked by avoidable cost, each with its score, top drivers, recommended action, cost, and approve or override buttons that call the decision endpoint.
- Tariff scenarios (trade compliance): a policy selector, a slider for the action date, and an input for the probability of enactment, which call /scenario/tariff to recalculate exposure and the probability at which acting breaks even.
- Network map (operations): Node_Lookup coordinates with active event areas and exposed lanes.
- Data quality (data owners): pass rates for each validation rule and the last successful load.

Each recommendation will show why it was made and be easy to correct, consistent with guidelines for human interaction with artificial intelligence (AI) (Amershi et al., 2019).

Preserving Integrity, Governance, and Access Control (2.5)

Integrity will be enforced by the database rather than by application code. Foreign keys will block orphan legs; CHECK constraints will keep flags at 0 or 1, tariff_rate between 0 and 1,

and delay = actual − planned; unique constraints will block duplicate keys; and timestamps will be stored as UTC types. The provided data already satisfy every one of these constraints. Loads will pass a validation gate, and failure on an integrity rule will stop the load. Every prediction will record the data and model versions it used, so any alert can be reproduced. Access will depend on role: planners will read the v_order_journey view and decide alerts, trade compliance alone will edit tariff events, finance alone will see the value and tariff detail of each order, and administrators will manage models. All read and write events will be logged and monitored.

Question 3: Value Measurement

Acting at Tariff Announcement Versus the Effective Date (3.1)

Table 3 splits the 470 shipments from China that paid the higher rate by departure timing. Incremental tariff is the amount paid above the 14.4% rate in force before the effective date:

$$\text{Incremental tariff} = \sum(\text{tariff_rate} - 14.4\%) \times \text{order_value_usd}$$

where tariff_rate is the rate each shipment paid and 14.4% is the rate, weighted by order value, on arrivals from China before May 17.

Table 3

Tariff Exposure on Shipments From China by Decision Cohort

Departure window	Orders	Tariff paid	Incremental	Addressable when acting at
At sea before announcement (April 12–18)	8	$132,616	$61,044	Neither date
During announcement window (April 20–May 14)	33	$1,126,655	$505,831	Announcement only
After the window (from May 15)	429	$12,755,386	$5,627,396	Both, through sourcing
Total	470	$14,014,657	$6,194,270	

Simulated on the 2025 orders, acting at the effective date would have left all 33 window departures locked in. Acting at the announcement would have let Aztec move 18 of them (ocean bookings sailing by May 7) to air. Planned China air transit is at most 8 days, so they would have cleared before the effective date. That would have avoided $253,825 of incremental tariff for a $19,327 air premium ($1,074 per order, the average gap between China air and ocean freight), a net saving of $234,498. The other 15 departed from May 9 onward, too late for air to help. Even if the rate had begun a day earlier, 16 shipments would have qualified, for a net saving of $189,044. Early action would also have given Aztec a head start of 26 days on qualifying suppliers outside China. If qualification takes about that long, starting at the announcement would have had a supplier ready by the effective date. Waiting would have left the 40 China orders that departed over the next 26 days (May 18 to June 10) at the higher rate, costing $458,334 in added tariffs before any price premium from the new supplier.

Acting early risks a false alarm, since announced tariffs can be delayed or withdrawn. By the cost rule in Section 2.2, acting at the announcement pays off whenever the probability of enactment exceeds the air premium divided by the avoided tariff, here 7.6%. Operationally, early action would have added 18 air bookings between April 20 and May 7, about one a day. Waiting costs nothing to execute, but once the effective date passes, an avoidable cost becomes unavoidable.

Early Detection, Decision Timing, and Outcomes (3.2)

Detection timing decides which options remain. At booking (T0), the median import order was 692 hours (28.8 days) from its promise, and at port arrival (T1), 56 hours remained. A reactive process learns of the failure only after the promise has passed. Both points add value:

- At T0, 632 orders (21.1%) were booked with a planned wait before departure longer than their buffer. All 632 missed the SLA, half of all misses, and the proposed system could flag them before the promise is sent. Simulated on the 2025 orders, adding the planned wait before departure to each promised date (24.3 hours on average for imports) would have raised the share of orders delivered on time from 57.8% to 73.8% with no transport cost. Shipments would arrive no sooner; promises would simply match what the schedule allows.
- At T1, the international delay is known and the shortfall can be computed. Switching to a faster domestic option only when the shortfall is recoverable would have moved 277 orders, recovered 262 commitments, and lifted the share delivered on time to 66.6% for $65,817, or $251 per recovered order.

Together, the two changes would have cut violations from 1,265 to 707 with only 87 switches costing $22,196, because the promise fix removes most of the need to expedite. This matches Tomlin's (2006) distinction between mitigation (the promise fix) and contingency (switching after a delay). Both depend on early warning and recovery, which reduce disruption severity (Craighead et al., 2007).

Domestic Routing Strategies and Service Performance (3.3)

Chosen in advance, the domestic strategy barely changed reliability (Table 4). Disruption rates ranged from 17.3% to 20.0%, and a disrupted leg lost 24 to 27 hours whatever the mode. FTL, the most expensive option, had the lowest share delivered on time (51.5%) because its orders carried the most international delay (2.8 days on average) and the most infeasible promises (22.4%). Every strategy delivered all feasible, undelayed orders on time.

Table 4

Domestic Strategy Performance in 2025

Strategy	Orders	Planned hours	Cost per order	On time	Disruption rate
Parcel_Expedited	334	23.5	$125	61.4%	18.6%
Parcel_Standard	780	47.9	$62	58.1%	19.4%
FTL	540	52.2	$831	51.5%	20.0%
LTL	709	72.3	$314	59.5%	19.3%
Intermodal_Rail_Truck	637	105.5	$612	59.2%	17.3%

Table 5 compares rerouting policies at port arrival. The simulation moved Parcel_Standard to Parcel_Expedited and LTL or Intermodal to FTL at Table 4 averages, assuming each order kept its observed domestic delay. The targeted policy captured most of the benefit, 8.8 of the 10.1 points available. Widening the trigger to every order with under 12 hours of buffer would have cost $184,519 more for 0.1 more points, and switching every eligible order would have cost $489,053 more for 1.3 more points.

Table 5

Rerouting Policies at Port Arrival Simulated on 2025 Orders

Policy	Switches	On time	Added cost	Cost per recovered order
No intervention (baseline)	0	57.8%	$0	n/a
Recoverable shortfall only	277	66.6%	$65,817	$251
Buffer under 12 hours	961	66.7%	$250,336	$917
Switch every eligible order	2,126	67.9%	$554,870	$1,778

Key Performance Indicators (3.4)

Table 6 defines each key performance indicator (KPI) against the schema. Projected values were simulated by applying the adjusted promise dates, switching at port arrival, and the announcement trigger to the 2025 orders. They measure service against the revised promises and will serve as pilot targets. Resilience is measured as absorption, the share of disrupted orders still delivered on time, and recovery as lateness among disrupted orders (Hosseini et al., 2019; Pettit et al., 2010; Ponomarov & Holcomb, 2009).

Table 6

KPI Definitions, Computation From the Schema, and Values

KPI	Computation	Baseline	Projected
Orders on time	AVG(on_time_delivery_flag)	57.8%	76.4%
SLA violations	COUNT(on_time_delivery_flag = 0); reduction versus 1,265	1,265	707 (−44.1%)
Mean lateness	AVG(late_hours_vs_promise)	46.3 h	37.1 h
Tariff exposure reduction	Avoided ÷ SUM((tariff_rate − 0.144) × order_value_usd), rows in the effective window	$6.19M	−4.1% ($253,825)
Net tariff benefit	Avoided tariff − air premium (freight = intl_total_cost_usd − tariff_cost_usd)	n/a	$234,498
Cost per recovered order	Added domestic_cost_usd ÷ orders moved to on time	n/a	$277
Disruption absorption	Share of ground truth disrupted orders with on_time_delivery_flag = 1	14.3%	28.7%
Recovery lateness	AVG(late_hours_vs_promise) for disrupted orders	134.5 h	113.4 h
Alert quality	T1 alert precision / recall against on_time_delivery_flag	n/a	99.7% / 92.5%

Note. Earlier sourcing could add up to $458,334 of avoided tariff (−11.5% in total) before supplier premiums, which the schema does not record.

Question 4: Professional Communication

Structuring the Executive Report (4.1)

Written Report Structure

The introduction states three findings and the assumptions behind them. The body follows the order in which Aztec would judge the system: what information it needs, how it works, and what value it creates. This section sets out the tradeoffs, and the Conclusion closes with a recommendation. Key figures are tied to numbered tables.

Flow, Visuals, and Persuasion

Tables serve as the visuals that carry the argument so an executive can scan rather than read. Persuasion rests on transparency. Assumptions are stated at the start, and weaknesses such as the assumed $1,000 miss cost are named before a reader can raise them. Each figure can be recomputed from the data and the stated assumptions, and each section is supported by cited research.

Framing the Tradeoffs

Table 7 separates contingency, paid only when disruption occurs, from redundancy, paid every day (Sheffi & Rice, 2005; Tomlin, 2006). The recommendations emphasize contingency first because it is inexpensive and reversible, and treat redundancy, such as completely avoiding China sourcing, as a decision for the next sourcing cycle, once the pilot shows how often announcements become effective (Simchi-Levi et al., 2014).

Table 7

Cost, Resilience, and Feasibility of the Options

Option	Cost	Resilience effect	Feasibility

Add the departure wait to promise dates	$0 direct; import quotes 0–48 h longer (24 h average)	73.8% of orders on time	High. Sales must accept later quotes.
Reroute late arrivals at the port	$65,817	66.6% of orders on time	High. Use existing carriers.
Fly shipments at tariff announcement	$19,327 air premium	$253,825 tariff rate increase avoided	Medium. Requires management approval and air capacity.
Expedite every eligible order	$554,870	67.9% of orders on time	High. Poor value.
Shift sourcing away from China	Different supplier prices (not in the data and too uncertain to assume)	Up to $458,334 by starting at the announcement	Medium. New suppliers need assessments, but existing suppliers could be leveraged.

Responsible AI Usage and Disclosure (4.2)

AI use in the proposed decision system must be disclosed and governed. Under the AI Risk Management Framework (National Institute of Standards and Technology, 2023), each model will have an owner and a model card stating its training period, test results, and limits (Mitchell et al., 2019). Every alert will show its score, drivers, model version, and data timestamp. Verification will have three layers: automated data tests before scoring, quarterly rechecks of thresholds against outcomes, and reconciliation of KPI figures to finance records and carrier invoices before they reach management. Human-in-the-loop rules will follow levels of automation (Parasuraman et al., 2000). Once the pilot shows precision above 95% at a given score, the system may automatically execute reversible actions under $100 at or above that score, such as a $63 parcel upgrade, while lower scores that still justify the action will go to planners. Planners will approve mode changes, and sourcing changes and tariff responses will require manager approval. Overrides will be logged with reasons and reviewed monthly, which protects against abandoning an algorithm after one visible error (Dietvorst et al., 2015) and against

trusting it without checking. The goal is high automation with high human control (Shneiderman, 2020).

AI used to prepare reports, such as this one, also requires disclosure, which the Appendix provides (McAdoo et al., 2025).

Conclusion

Aztec's disruption problem is both a timing problem and a definition problem. The analysis shows that half of all missed commitments were already determined at booking, that the domestic network had enough speed to recover 262 commitments when delays were detected at the port, and that waiting for tariff certainty cost $234,498 on 18 shipments in one policy cycle, with up to $458,334 more at stake in sourcing lead time. Resilience depends on visibility and timely response (Christopher & Peck, 2004; Tang, 2006), which here means analyzing the data at the moments when decisions are still open. I recommend adding the departure wait to promise dates, rerouting late arrivals at the port, and flying future shipments when a new tariff is announced.

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Appendix

Disclosure of Generative AI Use

I used Claude (Anthropic, 2026) to support the revision and enhancement of this document. I asked the tool to perform the following to my completed draft: (a) grade my final draft against the requirements specified within the report; (b) trim sections for repetitiveness and unnecessary jargon; (c) identify my references and check their bibliographic accuracy against publisher records; (d) check for any contradictory assumptions made; (e) revise any sections where I overexplain; (f) restructure language to meet professional standards; (g) restructure formatting and visuals; and (h) validate the APA format for accuracy.

I reviewed every enhancement and revised the text before submission. I take full responsibility for the content. Link to chat prompt: https://claude.ai/share/abf26ff0-acd5-46a2-b5e9-189195bfcd31.